

Panel MS-AR(1): common intercept and linear trend

1990–2019, excluding the US, Germany, Japan, and China

1 Model

Seasonally adjusted real GDP per worker, 2026-09-17 vintage, 1990Q1 through 2019Q4. United States, Germany, and Japan are excluded. China is not in this vintage.

$$y_{it} = a + g t + z_{it}, \tag{1}$$

$$s_{i,t+1} \sim \Pi(\cdot \mid s_{it}), \tag{2}$$

$$z_{i,t+1} = (1 - \rho(s_{i,t+1}))\mu(s_{i,t+1}) + \rho(s_{i,t+1})z_{it} + \sigma(s_{i,t+1})\varepsilon_{it}. \tag{3}$$

The regime s_{it} is the one that produced z_{it} . ε_{it} is the innovation. g is the within-country slope, so a country that enters later does not shift g through its level. a is then set so that $z_{it} = y_{it} - a - gt$ has mean zero. Three regimes. The MS-AR imposes $E[z] = 0$. σ and ρ switch, with $|\rho| < 0.99$. Standard errors are delta-method from a numerical Hessian of the cycle parameters. a and g have none.

2 Common intercept and linear trend, 1990–2019, ex US, Germany, Japan, China

2026-09-17 SA panel, 1990Q1–2019Q4, excluding the US, Germany, and Japan. $y_{it} = a + gt + z_{it}$, g the within-country slope, $E[z] = 0$. 71 countries, 6159 observations, 1990-Q1–2019-Q4. Fitted countries: 71. Observations: 6159. Log-likelihood: 16192.30. The within-country slope is $g = 0.0174$ and the intercept that zeros the pooled mean of z is $a = -5.2938$ (no standard errors). a is the intercept at the first date in the estimation sample. $z_{it} = y_{it} - a - gt$ is the cycle in the figure. Its average within-country slope is zero. The MS-AR is the law of z , with $E[z] = 0$.

Table 1: Regime AR(1), transition matrix, and ergodic π . Columns are regimes.

	(0)	(1)	(2)
μ	-0.8121 (0.0391)	-0.7770 (0.0630)	1.1616 (0.0144)
σ	0.0154 (0.0003)	0.0613 (0.0022)	0.0098 (0.0002)
ρ	0.9900 (0.0000)	0.9900 (0.0000)	0.9900 (0.0000)
Π_0	0.9532 (0.0064)	0.0458 (0.0064)	0.0010 (0.0019)
Π_1	0.1397 (0.0247)	0.7978 (0.0293)	0.0625 (0.0116)
Π_2	0.0006 (0.0021)	0.0229 (0.0039)	0.9765 (0.0037)
π	0.4441 (0.0183)	0.1470 (0.0148)	0.4088 (0.0105)

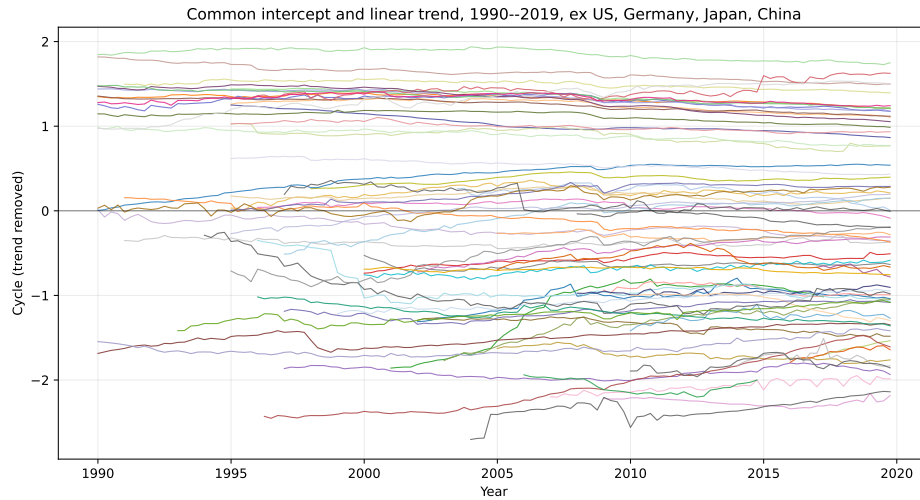


Figure 1: Country cycles $z_{it} = y_{it} - a - gt$.